

Borel sets — a refresher (v1)

Changelog (v0 → v1). Six corrections from review, plus three minor qualifications:

1. Borel-hierarchy notation is now the standard $\Sigma_\alpha^0/\Pi_\alpha^0$ (superscript **0**); superscript 1 is reserved for the projective hierarchy.
2. "Compact sets generate the Borel σ -algebra" is now restricted to **locally compact** second-countable spaces.
3. The convergence-set class is now $F_{\sigma\delta} = \Pi_3^0$ (was misstated as $G_{\delta\sigma} = \Sigma_3^0$); confirmed for both general continuous functions and power series on the circle of convergence.
4. The maximality statement about translation-invariant measures is corrected: the **Lebesgue** σ -algebra, not the Borel σ -algebra, is the maximal domain.
5. Regularity is now restricted to **Polish** spaces (inner regularity by compacts can fail on general metric spaces).
6. $\mathcal{S}'(\mathbb{R}^d)$ is now described as a **Suslin** space (not Polish, not metrizable).

The definition

For a topological space X , the **Borel σ -algebra** $\mathcal{B}(X)$ is the *smallest* σ -algebra containing all open sets:

$$\mathcal{B}(X) = \bigcap \{ \Sigma : \Sigma \text{ is a } \sigma\text{-algebra on } X, \tau_X \subseteq \Sigma \}$$

(the intersection of σ -algebras is again a σ -algebra, so this exists). Sets in it are **Borel sets**.

Two points worth internalizing:

- **Many equivalent generators.** On $\mathbb{R}^d$ (or any **locally compact** second-countable space), $\mathcal{B}(X)$ is equally well generated by: open sets, closed sets, compact sets, open intervals/boxes, half-lines $(-\infty, a]$. This is a standard π -system/Dynkin-lemma argument. **Caveat:** in a *general* second-countable space, compact sets need not generate the Borel σ -algebra — e.g. in an infinite-dimensional separable Hilbert space the closed unit ball is closed (hence Borel) but is not σ -compact, so the σ -algebra generated by compact sets is strictly smaller than $\mathcal{B}(X)$. The equivalence "open = closed = compact generators" is a feature of locally compact second-countable spaces, not of second-countable spaces in general.
- **Minimality is the point.** $\mathcal{B}(X)$ is the smallest σ -algebra making *every continuous function measurable*: if $f : X \rightarrow Y$ is continuous, $f^{-1}(U)$ is open, hence Borel. It's the default because analysis happens on topological spaces and uses continuous functions.

What's in it

By construction: complements, **countable** unions, countable intersections of Borel sets are Borel. That's it — uncountable operations are not allowed, and this is where everything breaks.

- $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ is Borel (each singleton is closed, in a T_1 space such as $\mathbb{R}$) — class F_σ (Σ_2^0).
- $\mathbb{R} \setminus \mathbb{Q}$ is Borel — class G_δ (Π_2^0).
- The Cantor set is closed (hence Borel), uncountable (cardinality $\mathfrak{c}$), and Lebesgue measure zero.
- A Vitali set is an uncountable union of Borel singletons and is **not** Borel — the first warning that countability in the definition is load-bearing.

The Borel hierarchy

Iterate countable unions/complements starting from open sets. Standard notation uses a **superscript 0** (superscript 1 is reserved for the *projective* hierarchy — analytic, coanalytic, etc.):

Class	Name	Description	Example
Σ_1^0	open	—	$(0, 1)$
Π_1^0	closed	—	$[0, 1]$
Σ_2^0	F_σ	countable $\bigcup$ of closed	$\mathbb{Q}$
Π_2^0	G_δ	countable $\bigcap$ of open	$\mathbb{R} \setminus \mathbb{Q}$
Σ_3^0	$G_{\delta\sigma}$	countable $\bigcup$ of G_δ	—
Π_3^0	$F_{\sigma\delta}$	countable $\bigcap$ of F_σ	convergence set of a sequence of continuous functions
...		continue trans- finitely up to ω_1	

Every Borel set of $\mathbb{R}$ appears at some level $< \omega_1$. Two facts:

1. **In metric spaces, closed sets are G_δ and open sets are F_σ :** $F = \bigcap_n \{x : d(x, F) < 1/n\}$ and $U = \bigcup_n \{x : d(x, X \setminus U) \geq 1/n\}$. This is special to metric spaces (it fails in general topological spaces). The inclusions are strict — $\mathbb{Q} \in \Sigma_2^0 \setminus \Sigma_1^0$ and $\mathbb{R} \setminus \mathbb{Q} \in \Pi_2^0 \setminus \Pi_1^0$ — so the hierarchy is already nontrivial at level 2.
2. **The hierarchy is strict** (up to ω_1 , on any uncountable Polish space), and Baire category is the clean proof: $\mathbb{Q}$ is F_σ but not G_δ — a dense G_δ in a complete metric space is comeager, while $\mathbb{Q}$ is meager. Dually, the irrationals are G_δ but not F_σ .

Historical anchor (and a nice test of the machinery): Borel's 1895 result that the set of convergence of a power series $\sum a_n x^n$ is Borel. Writing the Cauchy criterion $\forall k \exists N \forall m, n \geq N : |S_m(x) - S_n(x)| \leq 1/k$ and evaluating the levels — each $\{x : |S_m - S_n| \leq 1/k\}$ is **closed**, then $\bigcap_{m,n \geq N} (\text{closed})$ is closed, then $\bigcup_N$ is F_σ , then $\bigcap_k$ — gives a set of class $F_{\sigma\delta} = \Pi_3^0$. This is the same class as the convergence set of a general sequence of continuous functions, and the bound is sharp (Herzog-Piranian). One of the first deep uses of the hierarchy.

How many Borel sets are there?

On any **uncountable** second-countable space (so $\mathbb{R}, \mathbb{R}^d$, any uncountable separable metric space), $|\mathcal{B}(X)| = \mathfrak{c}$: every Borel set is built from a countable base by countable operations, and there are only $\mathfrak{c}$ such "Borel codes" (well-founded trees on a countable set). But $|\mathcal{P}(\mathbb{R})| = 2^\mathfrak{c} > \mathfrak{c}$. (For a *finite* second-countable space, $|\mathcal{B}(X)| = 2^{|X|}$, so the $\mathfrak{c}$ count is a feature of the uncountable case.)

Consequences:

- **Almost all subsets of $\mathbb{R}$ are not Borel.**
- Every set of cardinality $\leq \aleph_0$ is Borel (it's F_σ , provided singletons are closed — true in T_1 spaces such as $\mathbb{R}$).
- There are uncountable non-Borel sets (Vitali, Bernstein sets), all of cardinality $\mathfrak{c}$.

Borel vs. Lebesgue

The Lebesgue σ -algebra is the **completion** of $\mathcal{B}(\mathbb{R})$ w.r.t. Lebesgue measure:

$$\mathcal{L} = \{B \cup N : B \in \mathcal{B}(\mathbb{R}), N \subseteq M \text{ for some Borel null set } M\}$$

- Borel $\Rightarrow$ Lebesgue measurable; the converse fails. The Cantor set C is Borel and null; it has $2^{\mathfrak{c}}$ subsets, all Lebesgue measurable (measure 0), but only $\mathfrak{c}$ Borel. So most subsets of C are Lebesgue-measurable non-Borel.
- In L^p -theory the distinction is invisible (you work with equivalence classes mod null sets); it matters for pointwise-defined objects: versions of random variables, pointwise limits of measurable functions, etc.

Why this is “the” σ -algebra

1. **Maximality under countable additivity (Vitali).** There is **no** translation-invariant, countably-additive, normalized measure on the **full power set** $\mathcal{P}(\mathbb{R})$ — this is the content of Vitali's argument. The Lebesgue σ -algebra $\mathcal{L}$ (the completion of $\mathcal{B}(\mathbb{R})$) is the **maximal** σ -algebra (containing the Borel sets) on which such a measure exists: any σ -algebra $\Sigma \supseteq \mathcal{B}(\mathbb{R})$ supporting a translation-invariant, countably-additive measure finite on $[0, 1]$ must satisfy $\Sigma \subseteq \mathcal{L}$. Equivalently, Lebesgue measure is the *unique* translation-invariant, countably-additive measure on $\mathcal{L}$ with $\mu([0, 1]) = 1$. So $\mathcal{L}$ — not $\mathcal{B}(\mathbb{R})$ itself — is the largest sensible domain for the Lebesgue integral on $\mathbb{R}$.
2. **Regularity.** Every Borel probability measure on a **Polish** space is regular (Radon): outer regular by open sets, inner regular by compact sets. (Inner regularity by compact sets can fail on general metric spaces; the Polish hypothesis is what guarantees it.)
3. **Riesz-Markov-Kakutani.** Positive linear functionals on $C_c(X) \leftrightarrow$ Radon (Borel) measures. This is the bridge between “integrating against a measure” and “functionals on test functions” — a distribution is a continuous linear functional on test functions; a measure is a positive one.
4. **Bochner.** A continuous positive-definite function on $\mathbb{R}^d$ is the Fourier transform of a unique finite positive Borel measure — the characteristic function determines the distribution.
5. **Uniqueness by a determining class.** A probability measure on $(\mathbb{R}, \mathcal{B})$ is determined by its values on the half-lines $(-\infty, a]$ (Dynkin's π - λ lemma) — which is exactly why a CDF determines a law.

Standard Borel spaces

The Borel σ -algebra of a **Polish** space (separable + completely metrizable) is a *standard Borel space*. Two theorems make this the default universe of probability and analysis:

- **Borel isomorphism theorem:** any two uncountable standard Borel spaces are Borel-isomorphic (a bijection measurable in both directions). So as far as Borel structure goes, $\mathbb{R}$, $\mathbb{R}^d$, $[0, 1]$, Cantor space $\{0, 1\}^{\mathbb{N}}$, $C[0, 1]$, $L^p[0, 1]$ ($p < \infty$) are all “the same space.”
- **Measure isomorphism theorem:** any two atomless standard Borel probability spaces are measure-isomorphic. This is why all “continuous” random variables live in the same world, and why you can model any atomless law as a map into $([0, 1], \lambda)$.

Where it shows up in physics

1. **Random variables.** $X : (\Omega, \mathcal{F}, P) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a random variable iff $X^{-1}(B) \in \mathcal{F}$ for every Borel B . The law $P \circ X^{-1}$ is a Borel probability measure on $\mathbb{R}$, determined by the CDF. This is the single most common place the word "Borel" meets a physicist.
2. **Spectral theorem (QM).** For a self-adjoint A with spectrum $\sigma(A)$, there is a unique projection-valued measure $E : \mathcal{B}(\sigma(A)) \rightarrow \mathcal{P}(\mathcal{H})$ (countably additive in the strong operator topology) with $A = \int_{\sigma(A)} \lambda dE(\lambda)$. For each state ψ , $\mu_\psi(B) = \langle \psi, E(B)\psi \rangle$ is a Borel measure, and the **Borel functional calculus** gives $f(A) = \int f dE(\lambda)$ for Borel-measurable f . Here "Borel" is not decoration — the domain of the PVM is the Borel σ -algebra of the spectrum.
3. **Path integrals / Wiener measure.** The rigorous (Wick-rotated) path integral is the Wiener measure on $C[0, 1]$: a Borel probability measure on a **Polish** space with $\mathbb{E}[x_s x_t] = \min(s, t)$. Because $C[0, 1]$ is Polish and the coordinate evaluations separate points, the cylinder σ -algebra (finite time evaluations) **equals** the full Borel σ -algebra — this is the countable-product regime where the two coincide. Feynman-Kac then recovers the kernel of e^{-tH} . (The uncountable-product pitfall below does *not* apply to $C[0, 1]$; it applies to the formal product space $\mathbb{R}^{[0,1]}$.)
4. **Euclidean QFT (Osterwalder-Schrader).** A Schwinger function satisfying the axioms reconstructs a measure on $\mathcal{S}'(\mathbb{R}^d)$ — a measure on a **Suslin space** (the strong dual of the nuclear Fréchet space $\mathcal{S}(\mathbb{R}^d)$; not Polish, since it is not metrizable), handled by the general theory of Radon measures on such spaces.
5. **Statistical mechanics.** Gibbs measures on configuration space $S^{\mathbb{Z}^d}$ (Cantor space, compact metric) are Borel probability measures satisfying the DLR equations; the DLR equations fix the measure on cylinder sets, a determining class.
6. **Ergodic theory / statistical physics.** Measure-preserving transformations on standard Borel probability spaces; Birkhoff's ergodic theorem for L^1 observables; ergodicity = no nontrivial invariant Borel sets.
7. **Weak convergence of measures** (everywhere in PDE and analysis): Prokhorov — on a Polish space, tightness $\Leftrightarrow$ relative compactness in the weak topology; Portmanteau — $\mu_n \rightarrow \mu$ weakly iff $\int f d\mu_n \rightarrow \int f d\mu$ for bounded continuous f , equivalently $\mu_n(C) \rightarrow \mu(C)$ for every *continuity set* C (a Borel set whose boundary has μ -measure zero).

Pitfalls

- **Uncountable unions of Borel sets need not be Borel** — a Vitali set is an uncountable union of Borel singletons.
- **"Described" $\neq$ Borel.** Only countable operations are licensed. Definitions involving uncountable quantification over sets, or a choice (Vitali), typically land outside $\mathcal{B}(\mathbb{R})$.
- **Products.** For *countable* products of second-countable spaces, $\mathcal{B}(X^{\mathbb{N}})$ = the cylinder σ -algebra (e.g. Wiener measure on $C[0, 1]$, where coordinate evaluations separate points). For *uncountable* products (e.g. the formal path space $\mathbb{R}^{[0,1]}$), the cylinder σ -algebra is strictly smaller than the Borel σ -algebra of the product topology — this is the technical crux of formal measures on infinite-dimensional spaces.
- **Second countability is doing work.** On an uncountable discrete space, Borel = the full power set. The "c many Borel sets" fact is a feature of uncountable second-countable spaces.
- **Borel vs. Lebesgue** differ only by null sets on $\mathbb{R}^d$, but "Lebesgue measurable" is the right word for the domain of the Lebesgue integral; "Borel" is the topological notion.

Where the story continues

Above the Borel sets lies descriptive set theory: **analytic sets** (continuous images of Borel sets, equivalently projections of Borel sets in $\mathbb{R}^2$; these are the Σ_1^1 sets of the *projective*

hierarchy — note the superscript 1) are always universally measurable but not all Borel; then coanalytic sets (Π_1^1) and the rest of the projective hierarchy. The Borel sets are exactly the class closed under the countable operations that analysis actually uses — which is precisely why they, and not anything bigger, are the natural home of Lebesgue integration, probability laws, and spectral measures.